

# Antonio Cassio De Oliveira Segura

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## EDUCATION

### Purdue University, College of Engineering

*Bachelor of Science, Computer Engineering*

**2024 - 2028**

*West Lafayette, IN*

- **Coursework:** Advanced C Programming, Python for Data Science, Discrete Mathematics, Digital System Design, Multivariable Calculus, Differential Equations

## EXPERIENCE

### OneBlinC (BlinCAdvance)

*Software Engineer Intern*

**Jun 2026 - Present**

*Hybrid*

- Built serverless event-processing handlers on AWS (Lambda and SQS batch processing) with parameterized SQL, delivered across 3 merged PRs with unit tests and mock repositories, in a production AWS CDK stack (Python).
- Refactored 7+ event handlers into a unified dispatcher spanning the Django/DRF backend and Flutter mobile app, preserving backward compatibility and passing code review.
- Contributed across 3 production services (backend, mobile, and AWS infrastructure), improving error handling and CI checks to keep deployments reliable.

### The Data Mine · Purdue University

*Undergraduate Data Science Researcher*

**Aug 2025 - Dec 2025**

*West Lafayette, IN*

- Analyzed diagnostic trouble code (DTC) trends in transmission telematics data with Python and pandas, surfacing recurring failure patterns for the team's vehicle-performance insights.
- Built automated Python scripts for data engineering and trend detection across real-world telematics datasets, cutting manual analysis time for the research team.

### Purdue University College of Engineering

*Undergraduate Teaching Assistant, ENGR 13300*

**Aug 2025 - May 2026**

*Indianapolis, IN*

- Supported ENGR 13300 instruction by leading weekly in-class exercises and clarifying programming and engineering concepts using MATLAB, Python (pandas), and LaTeX; this hands-on guidance helped raise average exam scores and earned positive feedback from students

### Vertically Integrated Projects (VIP) · Allegion

*Undergraduate Researcher*

**Aug 2024 - May 2025**

*Indianapolis, IN*

- Programmed embedded C and Python firmware for IoT security prototypes and researched solar, thermal, and RF energy harvesting, analyzing energy-storage data in MATLAB to extend device battery life; collaborated with hardware and software teams using Onshape and PCB-design tools.

### Weldon School of Biomedical Engineering · Purdue

*Undergraduate Researcher*

**Jan 2025 - May 2025**

*Indianapolis, IN*

- Researched PEDOT:PSS-MXene neural bioelectrodes, running cyclic voltammetry, EIS, and pulse-waveform characterization and analyzing results in Python and MATLAB to confirm improved conductivity and stability (manuscript in progress).

## PROJECTS

### BoilerIndy [github.com/Antoniodoliveirasegura/boilerindy](https://github.com/Antoniodoliveirasegura/boilerindy)

**Mar 2026 - Present**

- Full-stack campus platform · React, Vite, Node.js/Express, Supabase (PostgreSQL), Playwright
- Engineered a full-stack campus app: 157 REST endpoints across 31 backend modules, 28 React pages, and a 31-table PostgreSQL schema, authoring ~80% of 197 commits while leading a 3-person team.
- Shipped 7+ feature areas (authentication, schedule sync with .ics calendar feeds, dining, community board, grade tracker, advertiser portal, AI assistant) and covered them with 24 unit and 6 Playwright E2E test files; deployed to Vercel and Render.

### WC2026 World Cup Forecast [github.com/Antoniodoliveirasegura/wc2026](https://github.com/Antoniodoliveirasegura/wc2026)

**Jun 2026 - Present**

- Statistical match & tournament model · Python
- Built a Dixon-Coles bivariate-Poisson scoreline model (maximum-likelihood fit, 3-year time decay) with an Elo cross-check and a connectivity-weighted squad-value prior, reaching out-of-sample RPS of 0.17, on par with betting-market accuracy.
- Simulated the full 48-team tournament over the real draw and host advantage, producing calibrated title odds where no nation exceeds a 16% championship probability.

### bysegura.com [bysegura.com](https://bysegura.com)

- Personal portfolio site · HTML / CSS / JavaScript, GitHub Pages
- Hand-built a no-framework responsive site with animated canvas backgrounds, persisted light/dark theming, and a live Twitch-status indicator served from a custom api.bysegura.com endpoint.

## TECHNICAL SKILLS

- **Languages:** Python, JavaScript (ES modules), C/C++, Dart, SQL, R, MATLAB, Bash
- **Frameworks & Libraries:** React, Node.js, Express, Django / DRF, Flutter, Vite, pandas, NumPy
- **Cloud & Data:** AWS (Lambda, SQS, CDK), Supabase, PostgreSQL, Vercel, Render
- **Testing & Tools:** Git / GitHub, GitHub Actions, Playwright, pytest, node:test, LaTeX